UGANDA CHRISTIAN UNIVERSITY
A Centre of Excellence in the Heart of Africa

FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY
DEPARTMENT OF COMPUTING AND TECHNOLOGY
EASTER 2026 SEMESTER EXAMINATION

PROGRAM: MSC DATA SCIENCE & MSC COMPUTER SCIENCE

YEAR: 1 & 2

SEMESTER: 1

COURSE CODE: DSC8307

COURSE NAME: DATA MINING, MODELLING AND ANALYTICS

EXAMINATION TYPE: PROJECT-BASED

EXAMINATION DATE: FEBRUARY - MARCH 2026

TIME ALLOWED: TWO WEEKS

Examination Instructions

1. The general Uganda Christian University examination guidelines and academic & financial policies apply to this examination. Violating any of the policies by the student automatically makes this examination attempt void, even if you have completed and submitted the answer booklet.
2. This exam consists of a project to be executed in 2 weeks.
   i. There are three parts to this project-based exam; Part A carries (30 Marks), Part B carries (40 Marks) and Part C carries (30 Marks).
   ii. To achieve all the parts, students should submit their work as python notebooks detailing the accomplishments for the different questions.
3. Every student has a responsibility to prove their contribution towards every milestone, and marks will be awarded to every student individually.
4. Answers to every question should start on a new paper within the examination answer booklet.
5. Uganda Christian University or its officials reserve the right of admission or disqualification from this exam.

PROJECT CONTEXT AND PROBLEM DESCRIPTION

Organizations across industries increasingly rely on integrated data systems capable of acquiring raw data, transforming it into analytical assets, generating predictive intelligence, and deploying models into operational environments.

For this examination, you are required to design and implement a data mining and analytics pipeline addressing a realistic decision-making problem within a domain of your choice.

You shall select **ONE** application domain:

- Climate Change : Predicting Extreme Events
- Retail: E-Commerce Analytics
- Healthcare: Patient Intelligence
- Financial: Risk & Customer Behavior Analytics
- Agriculture: Production & Forecasting
- Media: Text & Content Analytics

The selected problem must involve structured data and textual data components and must support predictive modeling and deployment considerations.

PART A: [TOTAL 30 MARKS]
Problem Framing, Data Mining, and Preparation

1. Based on the application domain selected, clearly define the analytical problem being addressed. The problem statement must identify the decision objective, relevant variables, and the role of data-driven modeling in solving the problem. **[10 MARKS]**

2. Based on the problem defined in Question 1, acquire or construct a dataset using publicly available data sources and/or data collected through web scraping. The dataset must contain variables suitable for predictive modeling. Provide a brief description of the dataset structure and characteristics. **[10 MARKS]**

3. Using the dataset obtained in Question 2, perform data cleaning and preparation to address missing values, inconsistencies, and outliers. Save the cleaned and transformed dataset as: **Your_LastName.csv** e.g, **Nuwe.csv**

[10 MARKS]

PART B: [TOTAL 40 MARKS]
Feature Engineering, Modeling, and Prediction

Use the transformed dataset generated in Part A that you saved as **Your_LastName.csv**.

1. Based on the problem being addressed, perform feature engineering and variable selection. Justify the relevance of selected features in relation to the predictive objective. **[10 MARKS]**

2. Develop machine learning models appropriate for the analytical task defined in Part A. At least two different algorithms must be implemented and their performance compared using suitable evaluation metrics. The best performing model shall be saved as: **Your_LastName_best_model.pkl or Your_LastName_best_model.joblib**, e.g, **Nuwe_best_model.pkl or Nuwe_best_model.joblib**

[15 MARKS]

3. Using the saved model from Part B (Question 2), generate predictions on a testing dataset or hold-out subset derived from the original data. Save the prediction output as:
Your_LastName_predictions.csv

[5 MARKS]

4. Interpret the performance of the developed models. Your discussion must include evaluation metrics, sources of prediction error, and practical implications of model inaccuracies within the selected domain. **[10 MARKS]**

PART C: [TOTAL 30 MARKS]
Text Mining, NLP, and Deployment Design

Continue using the problem selected in Part A.

1. Based on the domain and problem context, obtain or construct a textual dataset relevant to the analytical objective. Apply natural language processing techniques to extract meaningful insights. The analysis must include at least one visualization and one NLP-driven analytical task. **[10 MARKS]**

2. Load the predictive model saved in Part B **Your_LastName_best_model.pkl or Your_LastName_best_model.joblib** and demonstrate its reuse by generating example inferences. **[5 MARKS]**

3. Design a deployment architecture suitable for operationalizing the developed model using a framework of your choice (e.g., Streamlit, Flask, FastAPI, or Django). The design must clearly illustrate how users interact with the model and how predictions are produced. **[10 MARKS]**

4. Discuss ethical, privacy, and reliability considerations associated with deploying the proposed analytical system in a real-world environment. **[5 MARKS]**

Submission instructions:

- Python notebooks containing all analysis
- Clean dataset (Your_LastName.csv)
- Saved model (Your_LastName_best_model.pkl / joblib)
- Prediction outputs (predictions.csv)
- NLP outputs and visualizations
- Deployment script (Flask / Streamlit / Django)
- Final report (PDF, 4–6 pages) including the link to running application with the user guidelines.

Save and submit a zipped file of all your work in this naming convention format:
ACCESS NO_PROGRAM_COURSE & YEAR_COURSE UNIT NAME (e.g., A12345 or BBA3_Data mining, Modelling and Analytics). The submission of your examination answers should be made through the University through the e-learning platform (Moodle) before the cut-off date (19th March 2026).

◀ END OF EXAM GUIDELINES ▶